3. Model Development and Evaluation Report

3.1 Dataset Splitting

- **Train-Test Split Ratio:**
  - **80%** training dataset
  - **20%** testing dataset
- **Outputs:**
  - X_train, X_test
  - y_train, y_test

3.2 Models Implemented

1. **Logistic Regression**
   - Baseline linear classification model.
   - Highly interpretable with clear feature weight coefficients.
2. **Decision Tree**
   - Non-parametric tree-based classifier.
   - Effectively captures non-linear decision boundaries and easy to visualize.
3. **Random Forest**
   - Ensemble bagging model combining multiple decision trees.
   - Reduces variance and prevents overfitting.
4. **Gaussian Naive Bayes**
   - Probabilistic classifier based on Bayes' theorem assuming feature independence.
   - Highly efficient and performs well with scaled numerical inputs.
5. **K-Nearest Neighbors (KNN)**
   - Instance-based lazy learning classifier.
   - Classifies samples based on proximity in the feature space.

3.3 Model Training

- **Cross-Validation Setup:**
  - 5-Fold Stratified Cross-Validation (StratifiedKFold(n_splits=5, shuffle=True)) to preserve class proportions across folds.
- **Training Execution:**
  - Models trained using standard fit methodology (model.fit(X_train, y_train)).

3.4 Model Evaluation

Metrics Used (Classification):

- **Confusion Matrix**
- **Accuracy**
- **Precision**
- **Recall**
- **Misclassification Rate** *(calculated as 1 -Accuracy)*

3.5 Results Summary

Model	Accuracy	Precision	Recall
Logistic Regression	**88%**	**71%**	**54%**
Decision Tree	77%	33%	32%
Random Forest	85%	85%	17%

3.6 Model Selection

- **Selected Model: Logistic Regression**

Reason:

- **Highest Accuracy & Balance:** Achieved the highest overall cross-validation accuracy (~88%) and strong balanced performance across both precision and recall (yielding the highest overall F1 score).
- **Better Generalization:** Handles linear separation reliably on standardized features without overfitting compared to deeper tree or neighbor models.
- **Interpretability:** Provides transparent coefficient tracking for human resources insights into workforce attrition.